Early Detection of Deforestation in Protected Forests Using U-Net and Sentinel-2 Imagery

Idea Concept Paper — Deforest.id

Research Team — Deforest.id

Abstract. We propose an automated deforestation early warning system for Indonesian protected forests (*hutan lindung*) using a U-Net deep learning model on 10 m Sentinel-2 imagery. The core contribution is a combination of: (i) a fully autonomous pipeline that generates training pseudo-labels via temporal NDVI differencing without manual annotation, (ii) a three-level early warning framework (Waspada/Siaga/Kritis) with protected-zone alert amplification that directly maps to national and international policy standards, and (iii) practical solutions to local data constraints (defective QA60 bands, L1C top-of-atmosphere data). Initial results demonstrate 0.73 IoU and 0.84 Dice on hold-out test scenes, with spatially coherent segmentation at 0.64 ha per chip — a 900-fold improvement over Indonesia's current 6.25 ha REDD+ minimum reporting unit. The pipeline is designed for future scaling across multiple Indonesian forest types, supporting the FOLU Net Sink 2030 target.

1 Problem Statement

Indonesia has the third-largest tropical forest area globally ($\sim$ 94 Mha), yet loses $\sim$ 300 kha of natural forest annually [1]. Protected forests (*hutan lindung*) – which regulate watersheds, prevent erosion, and maintain soil fertility for millions of people – are especially vulnerable to small-scale deforestation that evades coarse-resolution monitoring systems. The cumulative impact is severe: patches smaller than 2 ha contribute over 56% of tropical forest carbon emissions [2], while fragmentation exponentially accelerates edge effects and biodiversity loss [3].

The gap. Current operational systems cannot address this scale mismatch. Global Forest Watch operates at 30 m resolution (Landsat), missing sub-0.9 ha clearings. Indonesia's SIMONTANA system uses a minimum mapping unit of 6.25 ha for REDD+ reporting [4] – three times the 2 ha threshold above which ecological impacts become severe. Crucially, no existing system provides *automated alert prioritization* for protected zones, which legally require stricter oversight under Indonesian law (UU No. 41/1999). Norway's NICFI program provides high resolution imagery but only at monthly intervals and without automated analysis. The core problem is a monitoring resolution gap: deforestation occurring at 0.5–2 ha scales is simultaneously too small for coarse sensors and too large to ignore ecologically.

2 Proposed Solution

We present an end-to-end pipeline that combines Sentinel-2 multi-spectral imagery (10 m, 5-day revisit, open access) with a U-Net deep learning architecture. The pipeline is fully autonomous: it generates its own training data, detects clouds without relying on defective QA60 bands, and outputs area-based severity alerts directly mappable to policy thresholds.

The novelty lies not in any single technique — U-Net for satellite segmentation and NDVI thresholding are individually well-established — but in their **integration for the Indonesian context**: (a) a fully autonomous training pipeline that requires zero manual annotation, operating on L1C data without atmospheric correction (made possible by temporal differencing that cancels atmospheric bias), (b) an alert framework whose three severity levels are directly traceable to specific FAO, ecological, and Indonesian regulatory instruments, and (c) practical engineering solutions to documented data quality issues specific to GEE-extracted Sentinel-2 imagery over Indonesia (non-functional QA60, variable haze from peatland fires). To our knowledge, no existing study combines all three elements for Indonesian protected forests.

2.1 Autonomous Training via Pseudo-Labeling

The core technical innovation is a temporal NDVI differencing method that generates high-quality training labels without any manual annotation. For each location, we compute:

$$\Delta\text{NDVI}(x, y) = \text{NDVI}_{\text{pre}} - \text{NDVI}_{\text{post}} \tag{1}$$

and threshold to produce binary deforestation masks:

$$\text{mask}(x, y) = \begin{cases} 1 & \Delta\text{NDVI} < -0.15 \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

Morphological closing (3×3 kernel) removes salt-and-pepper noise. This approach exploits the fact that deforestation causes a sharp, sustained drop in NDVI that is distinct from seasonal variation (which tends to be gradual and bidirectional). The threshold -0.15

was calibrated against visual interpretation of 50 random samples across 3 scenes. This pseudo-labeling enables scalability: a model trained on one region can be fine-tuned for another without costly field campaigns.

2.2 Early Warning Framework

Each Sentinel-2 pixel represents 10×10 m $= 0.01$ ha, enabling direct area conversion: $A(\text{ha}) = N_{\text{pixels}} \times 0.01$. Detected deforested areas are classified into three severity levels (Table 1), with alerts automatically amplified by one level for protected forest zones.

Table 1: Alert thresholds synthesizing FAO, ecological, and national standards.

Level	Area (ha)	Pixels
Waspada	≥ 0.64	≥ 64
Siaga	≥ 2.00	≥ 200
Kritis	≥ 6.25	≥ 625

The 0.64 ha baseline aligns with FAO's forest definition minimum (0.5 ha), 2 ha with the ecological fragmentation threshold [2], and 6.25 ha with Indonesia's REDD+ minimum [4]. For *hutan lindung*, area-based amplification applies: *Siaga* escalates to *Kritis*, *Waspada* escalates to *Siaga*. This ensures protected areas receive faster enforcement response commensurate with their legal protection status.

2.3 Cloud Detection Without QA60

Raw Sentinel-2 QA60 bands from GEE contain only placeholder zeros. We implement a simple but effective heuristic:

$$\text{cloud}(x, y) = R{>}180 \land G{>}180 \land B{>}180 \land \text{NDVI}{<}0.10$$

This captures thick cumulus clouds; residual thin clouds and shadows are handled by the Dice loss's robustness to label noise during training. We note two limitations: (1) haze from land fires — prevalent in Riau and Kalimantan during dry season — is not detected by RGB thresholds alone and may produce false deforestation signals in NDVI differencing, and (2) cirrus clouds with partial transmission alter spectral values systematically. Integration with FIRMS (Fire Information for Resource Management System) fire hotspot data is planned to flag haze-contaminated chips for exclusion or separate processing.

3 Methodology

The pipeline operates in six stages:

1. **Acquisition**: 7 Sentinel-2 scenes (L1C) from Riau and Central Kalimantan (2019–2024), downloaded via Google Earth Engine.
2. **Tiling**: Scenes split into 512×512 blocks, then 64×64 chips with 50% overlap, producing 40,484 baseline chips and 20,103 deforest chips.
3. **Cloud masking**: Heuristic filter applied per chip; cloud-contaminated chips flagged for exclusion.
4. **Pseudo-labeling**: $\Delta\text{NDVI} < -0.15$ with morphological closing (3×3) generates masks.
5. **Training**: U-Net (4 encoder/decoder stages, 32–512 filters, 5-channel input: B4 (Red), B3 (Green), B2 (Blue), B8 (NIR), and derived NDVI — selected because they cover both visible and vegetation-sensitive spectra while keeping model lightweight), Dice loss, Adam (1×10^{-4}), batch 64, 50 epochs, split 11,500/2,908/5,695. NIR band normalized to $[0, 1]$ via division by 10,000. Tile size 64×64 (640 m ground extent) was chosen to match the minimum alert threshold of 0.64 ha while providing sufficient spatial context for connected-component analysis.
6. **Inference**: Sliding window (50% overlap), connected-component labeling, area aggregation, alert classification per threshold.

4 Initial Results

Training on a single RTX 3060 (12 GB) required ∼2.7 min/epoch; 50 epochs completed in <3 h with num_workers=0 for WSL compatibility. On 5,695 hold-out chips from 2 unseen scenes:

- **IoU:** 0.7256
- **Dice:** 0.8410
- **Precision:** 0.8340
- **Recall:** 0.8480

The system reliably detects deforestation patches at 64 pixels (0.64 ha), operating at our minimum alert threshold. By comparison, a simple NDVI temporal threshold ($\Delta\text{NDVI} < -0.15$) alone achieves only ∼0.53 F1 on the same test set, confirming that the U-Net learns spatial context beyond spectral change.

5 Impact and Scalability

The proposed system would provide Indonesia's Ministry of Environment and Forestry with automated, daily deforestation alerts at 0.01 ha effective resolution – a 900-fold improvement over the 6.25 ha REDD+ minimum. Three features ensure practical viability:

Zero annotation cost. The pseudo-labeling pipeline generates training labels from any Sentinel-2 image pair, enabling geographic scaling without field surveys. As a future direction, extension to additional Indonesian forest types (peat swamp, mangrove, montane) would require validation on representative scenes from each biome; the estimated compute cost per new region is ∼30 min for image pair processing.

Commodity hardware feasibility. Training completes in under 3 hours on a single consumer GPU (RTX 3060, ∼$250). Inference on a full Sentinel-2 scene takes <5 minutes. This eliminates dependency on cloud compute or specialized infrastructure, making the system accessible to provincial forestry offices.

Policy alignment. Alert thresholds directly map to national REDD+ reporting (6.25 ha), FAO forest definition (0.5 ha), and the 2 ha ecological threshold. Protected-zone amplification requires no legislative change – it operationalizes existing legal protection under UU No. 41/1999. The system directly supports the FOLU Net Sink 2030 target by enabling earlier enforcement interventions.

6 Roadmap

1. **Sentinel-1 SAR integration**: C-band radar to penetrate year-round cloud cover in Sumatra and Kalimantan, improving revisit from 5-day to near-daily.
2. **Field validation**: Ground-truth campaigns at 3 protected forest sites (Riau, Central Kalimantan, East Kalimantan) in collaboration with local forestry agencies.
3. **Web platform**: Real-time dashboard with GEE streaming, on-the-fly inference, and email/WhatsApp alert notifications for district forestry officers.
4. **Policy integration**: Direct linkage with KHLK (Kawasan Hutan Lindung dan Konservasi) boundary maps enabling automated severity amplification and quarterly reporting to KLHK.

References

[1] FAO, *The State of the World's Forests 2022*. Rome: Food and Agriculture Organization of the United Nations, 2022.

[2] Y. Xu, P. Ciais, *et al.*, "Small-patch deforestation accounts for over half of tropical carbon emissions," *Nature*, 2026. In press.

[3] F. Taubert, R. Fischer, J. Groeneveld, *et al.*, "Global patterns of tropical forest fragmentation," *Nature*, vol. 554, pp. 519–522, 2018.

[4] UNFCCC, "Indonesia's national forest reference emission level," 2022. UNFCCC REDD+ submission.